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Project Title: SportsForecast

Abstract:

SportsForecast is an application that utilizes weather APIs and Groq API to provide feedback on optimal weather conditions for sports and hobbies. Reducing the risk of health issues, injuries, etc. Instead of guessing with intuition, athletes and coaches have a proper way to get accurate weather feedback designed primarily for chosen sports.

Introduction:

1.1 Problem Statement:

Outdoor sports are highly dependent on weather conditions yet most players, coaches, and event organizers use their intuition to decide whether it is appropriate to play based on a general forecast. Organizing and playing in hot weather, rain, lightning, high winds, and air pollution can result in canceled games and events, damaged equipment, and injuries and heat or lightning-related illness for players and spectators. Because different sports are affected by varying degrees based on weather conditions, a generic forecast cannot guide sport participants and spectators effectively. Understanding the appropriate weather to play a specific sport is paramount because it ensures safety, optimizes performance and enjoyment, eliminates wasted effort and time, and supports communities with activity planning in areas where weather conditions such as extreme heat and incessant rains in a place like Tacloban City, for example, prevent them from enjoying their hobbies.

The proposed sports weather app will address the identified problem by transforming the raw information from the weather into specific sports recommendations to help players, coaches,

and event planners make informed decisions. The effectiveness of the sports weather app will be determined using testing procedures such as unit tests for the recommendation algorithms, API tests to verify the accuracy of the data, comparisons of predicted and actual results, and usability tests to ascertain the validity and reliability of the information. The process of building the app will be done using the Agile development methodology that supports iterative design to ensure applicability and continuous improvement. The development will be done in sprints, beginning with the creation of a minimum viable product aka MVP for one or two sports and location and later adding more features and sports depending on the findings.

1.2 Objectives:

General Objective:

To design a sports weather app that assists players, coaches, and event planners in making decisions about weather-related conditions that affect the game by providing them with relevant information in the form of specific recommendations for the sport.

SMART Objectives:

- **Requirements Gathering {Specific}:** To determine the weather conditions [temperature, rainfall, wind, lightning, and air quality] that mostly impact the sports of football, basketball, and badminton by surveying athletes, coaches, and event organizers.
- **Recommendation Logic {Measurable}:** To create sport-specific weather thresholds that provide the sport with a Good, Caution, or Do Not Play designation, which will be verified by at least two coaches or sports safety references.
- **Weather Data Integration {Achievable}:** To integrate the weather API that will provide hourly forecasts for Tacloban City, with at least 95% of requests parsed and displayed correctly in the integration testing phase.

- **Provide Safety Alerts {Relevant}:** To develop alerts for severe weather conditions such as lightning and extreme heat to ensure that users do not engage in dangerous activities.
- **Have An Agile Delivery {Time-bound}:** To design and develop the app through 2-week sprints, deliver an MVP [Minimum Viable Product] in the middle of the development period, and have a pilot version by the end of the project.

Methodology:

2.1 Features

- **Real-Time Weather Tracking:** SportsForecast updates weather data every 90 seconds to capture sudden meteorological changes and supply up-to-date insights for users and secondary system features.
- **Live AI Weather Analysis:** Between tracking intervals, an AI engine analyzes the collected weather data to evaluate whether current conditions are optimal for the selected sport.
- **Smart Notifications:** Generates instant alerts when conditions become optimal for play, as well as real-time updates when weather conditions significantly improve or deteriorate.

2.2 Inputs & Outputs

Inputs:

User Configuration & Preferences:

- **Selected Sport**: User input indicating the target sport (Football, Basketball, or Badminton).
- **Venue Type / Environment**: Selection of indoor vs. outdoor setting (e.g., covered court vs. open field/court).

- **Location Data**: Automatic GPS coordinates or manual location entry (e.g., Tacloban City).
- **Notification Thresholds**: User-defined alert preferences (e.g., alert on condition changes, safety warnings only, or optimal playability).

Outputs:

Playability & Safety Status:

- **Sport Status Designation**: Color-coded playability status (Good / Optimal, Caution, or Do Not Play / Dangerous) generated by evaluating live metrics against sport-specific thresholds.
- **AI Weather Insights**: Natural language analysis generated by the Groq API explaining *why* conditions are favorable or dangerous (e.g., "High wind speed of 24 km/h will significantly disrupt shuttlecock trajectory during outdoor badminton").

System & User Alerts:

- **Push Notifications**: Instant pop-up alerts triggered when conditions transition to Optimal, or when weather rapidly deteriorates (e.g., sudden offset of heavy rain, high heat index, or flood risks).
- **Safety Warnings**: High-priority hazard advisories for extreme heat (risk of heat exhaustion) or severe weather events.

Visual Dashboard & Metrics Display:

- **Live Parameter Cards**: Clean UI display of parsed weather metrics (Temperature, Rain chance, Wind speed, Humidity, AQI).
- **Hourly Outlook / Trend Summary**: Visual indicator showing forecasted condition changes over the next few hours to assist in activity planning.

2.3 Logic Plan:

